Open feeds and curation power PH laptop picks

No single free source delivers Philippine laptop specs, PHP prices and benchmarks together, so a recommender has to combine five layers. **Specs** come from Open Icecat plus first-party OEM pages (Lenovo PSREF, Apple PH). **New PHP prices** come from the keyless Shopify `/products.json` feeds of PC Express, Octagon and Power Mac Center. **Used PHP prices** come from a depreciation model calibrated on TipidPC listings. **CPU/GPU performance** comes from Blender Open Data (CC0). A **hand-curated table** covers components and chassis material. The obvious shortcuts are all closed by licensing. Notebookcheck, TechPowerUp, Geekbench, PassMark and PugetBench forbid commercial reuse. Carousell and Facebook Marketplace forbid scraping, and under NPC Advisory 2026-01 that scraping now also carries Data Privacy Act exposure. On the buying side, **16 GB RAM is the 2026 floor for every use case**, set by Microsoft's Copilot+ minimum and the 16 GB base of the MacBook Air M5. For coding and school, CPU architecture decides more than speed because Android Studio, SolidWorks and Revit need Windows on x86. For gaming and 3D, GPU wattage and VRAM matter more than the GPU's name. Apple's M5 Pro/Max leads video and photo editing. Indicative PHP bands are **₱25–40k** for a capable 16 GB everyday laptop, **₱72–130k** for 1080p gaming, **₱160–250k** for 1440p gaming and serious 3D, and **₱250–440k** for flagship RTX 5080/5090 machines. A memory-driven price hike in June 2026 added ₱10–27k to Apple models, so static price lists go stale within months. That is the strongest argument for ingesting prices daily.

Part 1: Data sources

Only OEM pages and Open Icecat carry current, spec-rich laptop data

Free spec sources split into two groups: a few current, structured ones and many stale or thin ones. **Open Icecat** is the only free, structured, multi-brand catalog. Registration is free. A JSON endpoint (`live.icecat.biz/api`) takes an Icecat username plus `api-token` and `content-token` headers, and lookups use Brand + MPN, GTIN or Icecat ID. Responses include `GeneralInfo` (brand, GTINs, family/series), typed `FeaturesGroups` with raw and display values, images and `Variants` (Icecat JSON manual). The catch is coverage. Open Icecat includes **only brands that sponsor it** (iceclog). ASUS is a confirmed sponsor (iceclog ASUS Spain), but no current public sponsor list could be retrieved, so whether HP, Lenovo, Dell, Acer, MSI and Apple are free remains unknown until you register and query the brand list. The 2026 Fair Use Policy also matters. It **prohibits AI/ML training on Icecat data without a commercial license** and charges €0.30 per 1,000 datasheets beyond **1 million per month** per repository (Open Icecat Fair Use Policy). Images must be self-hosted, products with future release dates return 404 (Icecat JSON manual), and an older page requires a "Specs Icecat" attribution link (iceclog). Until the Open Content License text confirms otherwise, treat attribution as required.

First-party OEM documents are the most complete source and the only reliable one for chassis material. **Lenovo PSREF** covers products launching in Asia-Pacific, among other regions. It splits platform specs (PDF) from per-MTM configuration detail (online) (ThinkWiki PSREF). The ThinkPad X1 Carbon Gen 13

PDF, for example, lists battery Wh, per-CPU weight ("Starting at 1006 g") and "Case Material: Carbon fiber (top), magnesium or aluminium (bottom)" (PSREF PDF). PSREF has no documented API. An undocumented backend at `psrefapi.lenovo.com` answers `NewUpdated` queries by date range (psrefapi probe), but an undocumented endpoint can break without notice. **Apple's Philippine spec pages** are server-rendered and state material, metric weight and chip (Apple PH MacBook Air specs). Apple's robots.txt blocks only shop overlay paths (Apple robots.txt).

Everything else is too thin, stale, region-locked or legally awkward. **Wikidata** is CC0 but holds identity, not specs. A live query found only 11 laptop items dated 2023 or later and mass on just 34 items (Wikidata Query Service). The **Best Buy Products API** is US-only and its terms forbid caching "except on a temporary basis" (Best Buy API docs), which rules out building a persistent catalog from it. **UPCitemdb** allows 100 free lookups per day (UPCitemdb). **Open Products Facts** has 11 laptops in total (OPF). The Kaggle, Hugging Face and GitHub laptop datasets are snapshot price scrapes from the US, India, Spain, Algeria or Bangladesh, mostly unlicensed or stale (Baselight Kaggle mirror, HF bitsnaps/laptops_specs, GitHub search). They are useful as schema references and test fixtures, not as a live catalog.

The sites people usually scrape are mostly closed. Notebookcheck states "It is not allowed to copy any content" without permission and sits behind a Cloudflare challenge (Notebookcheck, robots.txt). GSMArena blocks ClaudeBot, GPTBot and other AI crawlers by name and covers only phones (GSMArena robots.txt). LaptopMedia's robots.txt is permissive and lists a sitemap (LaptopMedia robots.txt), but its spec database is a compiled work, so get written permission before reusing it. OEM sites generally allow product pages while disallowing their APIs: Dell and HP block `/api/` paths (Dell, HP), ASUS blocks spec-filter URLs but publishes sitemaps (ASUS), and MSI returns Akamai "Access Denied" to plain requests (MSI).

Spec source	Coverage	Freshness	Access and limits	License / constraint
Open Icecat	Sponsor brands only, full spec sheets, GTIN/MPN keyed	Current	Free account, tokens, one product per JSON call, XML index for bulk, 1M sheets/month	No AI training, self-host images, attribution likely
Lenovo PSREF	All Lenovo, incl. case material, weight, Wh, per-MTM configs	Current	Web/PDF, undocumented JSON API	Website terms not reviewed
Apple PH spec pages	~10 laptop SKUs/year, material, kg	Current	Server-rendered HTML	Store facts, not copied prose
Other OEM sites (ASUS, HP, Dell, Acer)	Own brand, PH regional pages show what is sold locally	Current	Sitemaps, `/api/` disallowed	Terms not reviewed
Wikidata	Brand and family names only	Effectively no 2025–26 models	Open SPARQL	CC0
Best Buy API	US assortment, dimensions, weight, free text	Current	Key by email	Temporary cache only
Kaggle / HF / GitHub sets	Foreign price scrapes	Mostly 2023–2025 snapshots	Download	Often unlicensed

Chassis material deserves its own note because no open aggregate carries it as a structured field. Material is a property of the model family, not the configuration. Fill it from OEM spec sheets first, then from an Icecat housing feature if one exists (unverified), then by manual curation with a `source_url` and `verified_at`. If none of those has it, show "not stated" rather than guessing. Store the lid and the bottom separately, because OEMs mix materials, as the X1 Carbon entry shows.

Shopify feeds give PHP shelf prices free, used prices must be modeled

New-retail PHP pricing has one excellent free source that most developers miss. Three major PH retailers run on Shopify and expose public `/products.json` feeds that need no key. On 2026-10-08, **PC Express** returned 250 products per page with price, `compare_at_price`, availability, SKU, vendor, `product_type` ("Laptops") and an `updated_at` timestamp from the same day (pcx.com.ph/products.json). **Octagon** and **Power Mac Center** expose the same feed, and Power Mac Center's SKUs are Apple part numbers such as MX532ZP/A (Octagon, Power Mac Center). All three robots.txt files declare public product HTML crawlable and ask only that checkout not be automated (pcx robots.txt). The feeds carry current price only, so price history has to be built from daily snapshots. Other chains allow crawling of clean product URLs through sitemaps: **Villman** allows everything (villman.com/robots.txt), **Silicon Valley** runs WooCommerce with a sitemap (robots.txt), and **Abenson** blocks only query and search paths (robots.txt). None of their terms of service were reviewed.

Marketplace prices from Shopee and Lazada are available only through affiliate programs that require approval. Shopee's Affiliate Open API offers product offer lists and short links (Shopee Help). It is GraphQL with HMAC signing, but the only documented endpoint is Brazil's (gregojoao/shopee-affiliate). A 2023 guide lists PH eligibility as 1,000+ social followers (Ginee PH). LazAffiliate exposes a CSV/TSV datafeed and an Open API behind its dashboard (Bryan Bonifacio), and Involve Asia offers advertiser datafeeds with name, price, URL and category (Involve Asia). The seller-side Lazada and Shopee Open Platform APIs return only your own shop's catalog (Lazada /products/get), so they are useless for a recommender. No PH price-comparison site offers an API. iPrice renders through JavaScript, and Pricebook's robots.txt disallows everything (iprice.ph, pricebook.ph). Affiliate APIs exist to generate promotional links, so pulling prices without promoting them may breach program terms. Read the terms after approval.

Second-hand pricing is where the legal risk sits. No PH used-laptop marketplace offers an API. **Carousell's** terms forbid "any robot, spider, scraper or other automated means" without express written permission (Carousell PH Terms), and **Facebook's** robots.txt prohibits automated collection without written permission (facebook.com/robots.txt). **NPC Advisory No. 2026-01** (13 April 2026) treats scraping of publicly available personal data as regulated processing under RA 10173. It rejects the idea that public availability equals consent and requires a Privacy Impact Assessment (Baker McKenzie). Scraping that violates a site's terms counts as "unauthorized", with criminal, civil and administrative liability (NDV Law). The advisory ties "unauthorized" to terms-of-service breaches, so third-party Apify actors for Carousell do not lower the risk. **TipidPC** is the exception: its 2026 robots.txt says listings "are public and meant to be found" and that category pagination reaches every live listing. It disallows `/api/`, search and filter parameters (tipidpc.com/robots.txt). Its terms still need review, and seller identifiers should be dropped at parse time.

The workable answer is a model anchored on the PH new price and calibrated by sampling. Buyback-vendor data puts MacBook value loss at 35/50/65% after one, two and three years and Windows at 45/65/78% (laptopreno.com). Premium ultrabooks lose 25–35% in year one and mid-range Windows laptops 40–50% (remoasset). Gaming laptops lose about 50% in the first year (growrk), and Macs with 16 GB+/512 GB+ retain 15–25% more value than base configurations (TechTimes). These figures come from vendors, conflict with one another, and none is PH-specific. A reasonable starting retention curve against PH SRP is Mac 70/55/42/32%, premium Windows 62/48/36/27%, mainstream Windows 55/40/28/20% and gaming 50/36/25/18% at years one to four. Recalibrate it quarterly against TipidPC data. eBay's Browse API is free at 5,000 calls/day (eBay), but it covers mostly US listings and reportedly prohibits computing average selling prices (eBay community). Back Market disallows all crawling (backmarket.com/robots.txt).

To turn TipidPC listings into a "typical used range", build the statistic in five steps. First, drop parts, defective, "LF/WTB/swap" and charger-only listings. Second, keep prices within 15–120% of the matching config's PH SRP. Third, bucket by condition and age. Fourth, remove outliers more than 3×1.4826×MAD from the median. Fifth, report P25–P75 as the range and the median as the "fair price", but only when at least five listings from the last 60 days exist. With fewer listings, fall back to the depreciation curve and label the figure "estimated". Asking prices run above sale prices, and no PH source quantifies the gap.

Blender Open Data is the only open benchmark that clears commercial use

Component specs have no open bulk feed. **Intel ARK** has no API, only a manual "Export comparison" to Excel for selected products (Intel support). **TechPowerUp's** robots.txt restricts its content to "personal, non-commercial use" and forbids scraping for datasets, software or any commercial purpose (techpowerup.com/robots.txt). That restriction carries over to the MIT- and Apache-licensed GPU packages derived from it, **dbgpu** (last release December 2025) and **RightNow GPU Database** (2,824 GPUs, last pushed January 2026) (PyPI dbgpu, HF gpu-database). Their licenses cover the packaging, not the underlying data. **Wikipedia** processor and GPU lists were edited within the past two weeks (List of AMD Ryzen processors, List of Nvidia GPUs), but they are CC BY-SA, so derived tables must carry attribution and share-alike. The defensible path is a hand-entered table of the roughly 150–300 CPUs and GPUs that matter (cores, boost clock, TDP, iGPU, VRAM, TGP range), taken from vendor pages, since individual facts are not copyrightable.

For benchmarks, **Blender Open Data** stands alone. Its data is **CC0** (about), and a daily snapshot (`opendata-latest.zip`, 101 MB zipped, 1.89 GB JSONL, dated 2026-10-08) records device name, backend (OptiX/CUDA/HIP/Metal/oneAPI/CPU), scene and samples per minute (snapshot). Its device strings also serve as a free registry of current chip names, including Core Ultra 3xx, Ryzen AI Max, Snapdragon X2, M5 variants and the RTX 50 Laptop parts. Coverage of 2026 laptop hardware is strong: **7,167 RTX 50 Laptop records**, 3,762 M5-family records and 2,316 Ryzen AI records. Core Ultra 300 (288) and Snapdragon X/X2 (327) remain thin. The medians below come from the snapshot's best-populated recent version (4.5.0, "monster" scene) and give a usable relative performance index.

Laptop GPU	Median samples/min	n		Laptop CPU	Median samples/min	n
RTX 5090 Laptop	3,837	153		Ryzen AI Max+ 395	222	64
RTX 4090 Laptop	3,420	70		Core Ultra 9 275HX	214	229
RTX 5080 Laptop	3,100	250		Ryzen AI 9 HX 370	136	76
M4 Max 40-core	2,514	124		Apple M5	132	48
RTX 5070 Laptop	1,910	231		Core Ultra 9 285H	108	54
RTX 5060 Laptop	1,595	302		Core Ultra 5 338H	98	5
RTX 5050 Laptop	1,314	145		Ryzen AI 7 350	88	59
M4 Pro 20-core	1,244	123		Core Ultra 7 258V	62	35
M5 10-core	833	33		Core Ultra 5 226V	50	10
Arc B390	602	16				

Source: Blender Open Data snapshot, version 4.5.0. Scores are not comparable across Blender versions, so pin one version.

Every other well-known benchmark database is off-limits for a commercial app. The **Geekbench 6** EULA is "personal, non-exclusive, non-commercial", bans automated access, and reserves all rights in results to Primate Labs (Geekbench EULA). **PassMark** forbids reproduction "for general publication or commercial purposes" (PassMark). **PugetBench**, the best proxy for video and photo work, requires a commercial license listed at $2,000 per technician (PugetBench licensing). OpenBenchmarking.org states no data license and blocked automated access (Phoronix Test Suite). No current open laptop FPS dataset exists. The only CC BY FPS set, OpenML 44992, dates from 2022 and is desktop-oriented (OpenML). No open battery-runtime dataset exists either.

Blender medians also hide a real weakness. Records carry no TGP, power mode or laptop model (snapshot), yet the same GPU name varies widely by power limit. Notebookcheck documents **65–130 W** for the RTX 3060 Laptop (Notebookcheck TGP list), and ASUS publishes per-model TGP tables for its 2025 ROG and TUF lines (ASUS ROG). The app should therefore store a per-laptop `gpu_tgp_w` from OEM spec sheets and scale the per-GPU median by it. Then map the result to labels such as "1080p High 60 fps class" and anchor those labels with a small, attributed set of review figures.

The recommended stack runs on five layers and two keys

The stack below uses only sources that are free and legally clear, or that become clear once an affiliate program approves you. Each layer has one job, and every stored field records its `source`, `source_url` and `fetched_at`.

Layer	Source	Cadence	Join key	Notes
Specs (multi-brand)	Open Icecat XML index + JSON	Weekly	Brand + uppercase MPN, GTIN secondary	Confirm sponsor brands after sign-up

Layer	Source	Cadence	Join key	Notes
Specs (gaps, material, weight, TGP)	PSREF, Apple PH, OEM `/ph/` pages via sitemaps	Weekly / on launch	MPN, Lenovo MTM	Polite rate limit, own UA
New PHP price	PCX, Octagon, Power Mac Center `/products.json`	Daily	SKU / MPN parsed from title	Snapshot daily for history
New PHP price (marketplace)	Shopee / Lazada affiliate APIs, Involve Asia feeds	Daily once approved	Title parse to config	Official/mall stores only
Used PHP price	Depreciation curve + TipidPC category crawl	Weekly crawl, quarterly recalibration	Normalized config	Drop seller data at parse
Performance	Blender Open Data snapshot	Monthly	Normalized CPU/GPU name	Pin Blender version
Components + material	Hand-curated tables from vendor pages	On chip/model launch	CPU/GPU string, model family	~150–300 chips, few hundred families/yr

The data model has two levels. A **model family** (brand + series + generation, such as "ThinkPad X1 Carbon Gen 13") holds the chassis attributes: material (lid and bottom separately), dimensions, base weight, ports, battery Wh and release year. A **config** is keyed by brand + MPN, with GTIN as a secondary key, and holds CPU, GPU, TGP, RAM, storage and panel. Lenovo MTMs make the split easy: the first four characters identify the family and all ten identify the config. Icecat's `family/series` field or a curated alias table handles other brands. SKUs that differ only in colour, OS, keyboard language or regional suffix collapse into one config for recommendations, and each retailer listing attaches to that config as an offer. The ingestion run follows seven steps. (1) Pull the Icecat index of new and changed laptops. (2) Fetch JSON per changed product. (3) Crawl OEM sitemaps for brands Icecat lacks. (4) Normalize units (mm, g, Wh, inches, nits). (5) Match and deduplicate. (6) Join CPU/GPU strings to the component and Blender tables. (7) Route null fields, chiefly material and TGP, to a curation queue.

Three legal guardrails keep the pipeline clean. First, never scrape Carousell or Facebook. Calibrate the used-price model with human spot-checks instead. Second, apply a field allowlist at parse time (title, specs, price, condition, city, date), publish only aggregates with n ≥ 5, keep raw listings for about 30 days, and write a lightweight PIA as the NPC advisory expects. Third, keep Icecat-sourced fields out of any model training, since the Fair Use Policy forbids it. Running a recommendation engine over the data is not training.

Part 2: Spec tiers by use case

16 GB is the floor, and architecture beats speed for school and coding

The 2026 floor rose for reasons outside the buyer's control. Microsoft's Copilot+ PC spec requires **16 GB RAM, 256 GB storage and a 40+ TOPS NPU** (Microsoft Support), and every MacBook Air M5 ships with **16 GB and 512 GB** as standard (GadgetPilipinas). About 70% of 2026 ultrabooks solder their RAM (Evet

ech), so an 8 GB budget unit stays 8 GB for life, and the app should flag it as a pitfall. Battery life no longer depends much on platform. Notebookcheck measured a Snapdragon X2 Elite Zenbook A14 at about **16 h** of Wi-Fi browsing, "pretty much identical" to the MacBook Air M5 (Notebookcheck), and Tom's Guide clocked a Panther Lake Dell XPS 14 at **20 h 41 m** (Tom's Guide). Long battery life also helps during the rotational brownouts that follow NGCP grid alerts in hot months (Inquirer).

Coding and engineering school depend on CPU architecture more than on performance. Android Studio on Windows supports **x86_64 only** (Android Developers), and Google has "no immediate plans" for an emulator on Windows ARM64 (Issue Tracker). Hyper-V on Windows ARM runs only ARM guests (Faceofit), while Docker Desktop does support Windows on ARM (Docker). Xcode 26 requires macOS, and a 24 GB Mac showed memory pressure with Xcode plus one or two simulators (Apple Developer Forums). SolidWorks 2026 needs x86-64 with 16 GB minimum and 32 GB recommended (Applicad) and does not support ARM (SolidWorks Help). Revit 2026 is Windows-only at 16 GB minimum and 32 GB recommended, and Enscape path tracing needs NVIDIA RTX (MyArchitectAI). Philippine schools publish almost nothing current to steer buyers. Mapúa's only online spec page dates from about 2012, and DLSU publishes only its own AY 2022–23 institutional unit spec (Mapúa Library, DLSU ITS). The app should ask for the student's program rather than assume a generic "student" profile.

The PHP anchors are well documented at the low end. Entry units such as the Vivobook Go 15 (8 GB) sell for ₱15,999–19,499, mainstream 16 GB units such as the Vivobook 15 and Acer Swift Go 14 for ₱22,999–36,999, and the Vivobook Pro 15 OLED for ₱39,999–49,999 (Tipidnation, Atome). The MacBook Air M5 launched at **₱72,990** for the 13-inch (₱65,990 education) and ₱85,990 for the 15-inch (GadgetPilipinas). The October 2026 Zenbook 14 runs from ₱76,995 (Snapdragon) to ₱129,995 (Core Ultra 9 386H) (Gadget Pilipinas), and an RTX 4050 Gigabyte A16 sells for ₱66,995 (Makotek). In the tables below, bands without a cited model are interpolations.

Use case	Minimum	Recommended	Ideal
Day-to-day / office	Any 2024+ CPU, 16 GB, 256–512 GB SSD, 1080p IPS. **₱25–40k**	Core Ultra 200V/300, Ryzen AI 300, Snapdragon X or Apple M, 16 GB / 512 GB, 15 h battery. **₱40–75k**	MacBook Air M5, X2 Elite or Panther Lake ultrabook, 16–32 GB, 1 TB, OLED, 16–20 h. **₱73–107k**
Coding	x86 Windows/Linux or Apple silicon, 6+ performance cores, 16 GB / 512 GB. **₱30–50k**	32 GB / 1 TB for Docker stacks, VMs or Android emulators. Mac with 24 GB+ for iOS. **₱75–140k**	32–64 GB, 1–2 TB, Core Ultra 9 H / Ryzen AI 9 or M5 Pro. Add an RTX 5070 Ti+ for CUDA ML. **₱140k+**
Business travel	≤1.5 kg, 16 GB / 512 GB, 12 h+ battery, USB-C PD charging. **₱40–75k**	≤1.3 kg, 14-inch 16:10, 15 h+, IR camera or fingerprint, 3-year international warranty. **₱75–130k**	About 1 kg (X1 Carbon Gen 13 class), 20 h, 32 GB, optional 5G. **₱130k+** (PH price unverified)
Music production	16 GB / 512 GB NVMe, low DPC latency on Windows. **₱40–75k**	32 GB / 1 TB. **₱75–140k**	64 GB+, 2 TB plus external SSD for sample libraries. **₱150k+** (unverified)

School level / program	Minimum	Recommended	Platform rule
Elementary to SHS	8 GB / 256 GB, ₱15–25k	16 GB / 512 GB, ₱25–40k	Any

School level / program	Minimum	Recommended	Platform rule
College, non-technical (business, nursing, humanities)	16 GB / 512 GB, ₱25–40k	Thin-and-light with 15 h battery, ₱40–75k	Any, but check exam lockdown-browser support for Mac/ARM
IT / CS	16 GB, ₱30–50k	32 GB, ₱75k+	x86 Windows or Mac. Avoid Snapdragon for Android work
Engineering (AutoCAD, SolidWorks, MATLAB)	16 GB, RTX 4050/5050 class, 512 GB, ₱67–80k	32 GB, 1 TB, ₱95–120k	Windows x86 only. No Mac or ARM
Architecture (Revit, Lumion, Enscape)	16 GB, RTX with 8 GB VRAM, ₱80–110k	32 GB, 1 TB, RTX 5060/5070, ₱110–160k	Windows x86 + NVIDIA RTX

Several caveats sit behind these tables. Travel weight targets come from a guide that sets sub-1.5 kg as the threshold and under 1.3 kg as the sweet spot (Evetech). MIL-STD-810H is a test method that brands self-run in varying subsets, not a certification (ASUS FAQ). Snapdragon's emulation layer blocks kernel-mode drivers, which breaks some corporate VPNs (Faceofit), and that is the main business-travel caveat. For music, 16 GB is the "absolute floor" and 32 GB is comfortable for heavy sample libraries (Beatsyncpro). On Windows, driver-induced DPC latency causes audio dropouts, so testing with LatencyMon matters more than CPU tier (Alesis). Warranty weighs more in the PH than spec sheets suggest. Since June 2026, ASUS gives Zenbook, ProArt and ROG units bought from authorized PH retailers a 3-year local and international warranty, backed by 100+ service centres (Unbox.ph), while Huawei honours warranty only for official-channel units (Huawei PH). The app should mark grey-market offers.

Gaming tiers hinge on GPU wattage and VRAM, not GPU name

The 2026 laptop GPU ladder runs Arc B390 iGPU < Radeon 8060S iGPU ≈ RTX 5050 ≈ RTX 4060 Laptop < RTX 5060 ≈ RTX 4070 Laptop < RTX 5070 < RTX 5070 Ti < RTX 5080 < RTX 5090. Notebookcheck expects the RTX 5050 to perform like the old RTX 4060 Laptop (Notebookcheck) and found the RTX 4070 Laptop "still slightly faster" than the 5060 in games (Notebookcheck). The 5050, 5060 and original 5070 all carry **8 GB**. NVIDIA added a **12 GB RTX 5070 Laptop** in April 2026 with identical cores and bandwidth, and it is sold under the same name as the 8 GB version (HotHardware). The recommender must therefore store VRAM per config, not per GPU name. On desktop RTX 5060 data, 8 GB costs 11% overall versus 16 GB at 1080p Ultra and 26% in ray-traced titles, and Spider-Man 2 exceeds 8 GB even at 1080p Very High (Newegg). The 5070 Ti to 5080 step is small: at an equal 110 W the 5080 led by only 13% (Notebookcheck DE). That makes the 5070 Ti the value point of the high tier, with the 5080 mainly buying 16 GB of VRAM.

Wattage changes the picture within each GPU name. GameStar measured an RTX 5070 Laptop **9% slower at 75 W and 32% slower at 50 W** than at 100 W in Cyberpunk 2077 (GameStar), and a 65–80 W RTX 5060 runs about 25% behind a 115 W unit (Evetech). A full-power 5060 can therefore beat a starved 5070. A MUX switch or Advanced Optimus adds 5–15% depending on the game (GameStar), and LaptopMag measured 14% in DiRT 5 but only 2% in Forza Horizon 5 (LaptopMag). Among CPUs, the Ryzen 9 9955HX3D is Notebookcheck's "best choice for gaming" (Notebookcheck), though the CPU matters little below the RTX 5070 Ti tier. Battlefield 6 asks for **16 GB dual-channel** RAM and 8 GB VRAM (EA), and Black Ops 7 installs at about 116 GB (KitGuru). That makes 1 TB the practical storage minimum. Some PH listings still ship single-stick 8 GB, such as the MSI Cyborg 15 C13WE0 at ₱99,990

(VillMan), and these deserve a warning. GTA VI has no PC release date (Gameluster), so the app should not size anyone's purchase for it.

Tier	GPU and TGP	CPU	RAM / storage	Display	PHP new (examples)
Minimum: esports (Valorant, CS2, Dota 2, Genshin)	RTX 5050 / 4060 at 95 W+, or Radeon 8060S	6–8 core H/HX	16 GB dual-channel, 512 GB–1 TB	1080p 144–165 Hz	**₱72–100k**: Cyborg 15 ₱71,995, LOQ 15 ₱79,995–109,995, Nitro V ₱76,999–95,995
Recommended: 1080p AAA	RTX 5060 at 100–115 W, MUX	Core i7 HX / Ryzen 7 HX	16–32 GB, 1 TB, spare M.2	1080p/1200p 165 Hz	**₱100–130k**: TUF F16 RTX 5060 ₱106,995, Legion 5 RTX 5050 ₱115,995
Recommended: 1440p AAA	RTX 5070 12 GB at 100 W+ or RTX 5070 Ti	Core Ultra 9 HX / Ryzen 9 HX	32 GB, 1–2 TB	2.5K 165–240 Hz	**₱160–250k**: Helios Neo Slim 16 ₱164,999, Strix G18 5070 Ti ₱179,995–244,995, Legion Pro 7 ₱249,995
Ideal: enthusiast	RTX 5080 / 5090 at 150–175 W	9955HX3D or Core Ultra 9 290HX Plus	32 GB+, 2 TB+	2.5K 300 Hz or 4K Mini LED	**₱250–440k**: Strix G16 RTX 5080 ₱319,995, SCAR 18 RTX 5090 ₱439,995

Prices come from PC Express, VillMan, PCX RTX 5060, PCX RTX 5070 Ti and REVU Philippines. Some PCX and VillMan listings are undated and may predate the 2026 memory-driven price rises. No PH price for the 12 GB RTX 5070 surfaced. Used RTX 4060 laptops have a ceiling set by their 2024 new prices of ₱65,999–75,495 (Unbox.ph), and they must undercut a new RTX 5050 clearly to justify losing warranty. PH room temperatures cut cooling headroom (Evetech), which favours thick 16–18-inch chassis with high sustained TGP over thin designs. No PH-specific thermal test exists.

Creative work splits between CUDA rendering and Apple's media engines

Creative work divides cleanly by workload. For **3D rendering**, VRAM sets the hard ceiling and NVIDIA keeps the broadest renderer support. An RTX 5090 laptop finished a Blender render in 11 s versus 16 s for a 16-inch MacBook Pro M5 Max (NoobFeed), and the Blender Open Data medians above put the 5090 Laptop at 3,837 samples/min against 1,910 for the 5070. Notebookcheck called the RTX 5070 Ti "High-End with just 12 GB VRAM" and measured it 23% ahead of the 5070 (Notebookcheck). Third-party guides put Blender's recommended spec at 8 cores, 32 GB RAM and 8 GB VRAM (RenderAHouse). For **video**, the picture reverses. In PugetBench Resolve the M5 Max scored about 127,090 against 85,114 for an RTX 5090 ProArt P16, and the M5 Pro roughly matched the 5090 (Notebookcheck). The M5 Max also exported HEVC faster (26 s vs 29.5 s) and kept working on battery while the RTX laptop shut down (NoobFeed). RTX 50 GPUs gained hardware 10-bit 4:2:2 decode in Premiere 25.3 and Resolve 20 (Puget Systems), and Resolve 20 on Windows needs 16 GB minimum, or 32 GB with Fusion (Blackmagic forum). For **photo work**, the M5 Max ran Lightroom 40% faster than the RTX 5090 laptop, and 58% faster on battery (NoobFeed), so single-thread CPU, RAM and panel accuracy matter more than the GPU. For **local AI**, memory capacity decides which models fit. At Q4 quantization, 8 GB runs 7–8B models, 16 GB

runs 14–20B, 24 GB runs 30–32B, and 70B exceeds every laptop GPU (AICybr). An RTX 5090 laptop generated 158 tok/s against 107 for the M5 Max (NoobFeed), but unified memory up to 128 GB lets Macs and the Ryzen AI Max+ 395 run models no laptop GPU can hold.

Workload	Minimum	Recommended	Ideal
3D rendering	RTX 5060/5070 8 GB at 100 W+, 32 GB, 8 cores, 1 TB	RTX 5070 Ti 12 GB or 5080 16 GB at ~140 W+, 32–64 GB, 2 TB, 16-inch vapor chamber	RTX 5090 24 GB at 150–175 W, 64 GB, 4 TB. M5 Max 64–128 GB when portability and scene size matter
Video editing	8-core CPU with hardware decode (Core Ultra H, M5, RTX 5050/5060), 16 GB, 1 TB, 100% sRGB	RTX 5070/5070 Ti or M5 Pro, 32 GB, 2 TB Gen4+, P3 OLED/mini-LED, Thunderbolt, SD reader	M5 Max 64 GB+ or RTX 5080/5090 with 64 GB, 4 TB, calibrated P3 panel
Photo editing	Core Ultra 5/7, Ryzen AI 5/7 or M5, 16 GB, 512 GB, sRGB panel with ΔE < 2	32 GB, 1 TB, P3 OLED/mini-LED with calibration report	64 GB, 2 TB, wide-gamut panel plus hardware calibrator
Local AI / ML	RTX 5060/5070 8 GB + 32 GB RAM, or Mac with 24–32 GB	RTX 5080 16 GB + 64 GB, or M5 Pro/Max 48–64 GB	RTX 5090 24 GB + 64–96 GB, or M5 Max / Ryzen AI Max+ 395 with 128 GB

Creative PHP bands sit higher than mainstream bands, and they moved up in 2026. Apple raised PH prices on 25 June 2026, citing an AI-driven memory shortage. The M5 MacBook Pro 14 16 GB/1 TB went from ₱112,990 to **₱139,990**, and the MacBook Neo rose ₱10,000 to ₱49,990 (Unbox.ph). At launch, the M5 MacBook Pro 14 started at ₱99,990 for 16 GB/512 GB and ₱125,990 for 24 GB/1 TB (NotypiGeeks). The 2026 Zenbook S14 and S16 sit at ₱129,995–134,995 and the Zenbook DUO at ₱179,995 (Newsbytes). RTX 5070 Ti laptops run ₱164,999–249,995 (PCX), and the RTX 5080 ProArt P16 launched at **₱289,995** (Technobaboy). That gives working bands of **₱75–130k** for entry creative work (Zenbook OLED, base M5 MacBook Pro, RTX 5050/5060), **₱130–225k** for mid-range (RTX 5070 Ti, Zenbook DUO) and **₱225–300k+** for high-end (ProArt P16, Legion Pro 7). No PH prices were found for the M5 Pro/Max MacBook Pro or RTX 5090 creator laptops. US pricing from $2,199 for the M5 Pro (PhoneArena) suggests roughly ₱140–170k, which is unverified.

Conclusion

Most of the engineering work is in licensing, not code. The best spec, benchmark and used-price data sits behind terms that forbid commercial reuse, and since NPC Advisory 2026-01, scraping in breach of those terms is also a privacy-law problem. The defensible build accepts two kinds of incompleteness. Used prices are modelled estimates, not scraped truth, and performance is a relative index from one CC0 benchmark scaled by TGP, not measured FPS. The app should label both honestly ("estimated", "relative"), because a recommender that cites its sources earns more trust than one that hides interpolation. Three unknowns gate the build and should be resolved first, since each determines how much manual curation the catalog will need: which brands Open Icecat covers for free, whether Shopee and Lazada affiliate terms allow price-only use, and what TipidPC's terms say.

On the buying side, the 2026 market punishes spec sheets that name a part without its context. The same "RTX 5070" can mean 8 or 12 GB and 50 or 125 W, the same "16 GB" can be soldered for life, and the same Windows laptop can or cannot run Android Studio or SolidWorks depending on whether it is

ARM. A recommender that asks two extra questions (school program or dev stack, and target resolution) and stores three extra fields (TGP, VRAM per config, CPU architecture) will give better advice than one that ranks laptops by benchmark score alone.